

Linux Troubleshooting Guide

1. Linux Basics

Linux servers are commonly used for cloud workloads, application servers, Docker hosts, monitoring systems, and Kubernetes nodes.

Basic system information can be collected with:

2. CPU Investigation

CPU alerts should be investigated by identifying the processes consuming CPU.

```
top
ps aux --sort=-%cpu | head -20
nproc
lscpu
```

Load average is different from CPU percentage and should be interpreted relative to the number of CPU cores.

3. Process Investigation

Every running process has a PID. Before terminating a process, determine what it is and how it was started.

```
ps -fp <PID>
ps -o pid,ppid,user,%cpu,%mem,etime,stat,cmd -p <PID>
pstree -p <PID>
```

The parent process matters because a supervisor may automatically recreate a terminated process.

4. Memory and Disk

Check memory and swap with:

Check disk capacity and inodes with: CMD: `df -h df -i du -sh /var/*`

Check disk capacity and inodes with:

Do not blindly delete files when disk usage is high. First determine what is consuming the space.

5. Services and Logs

systemd commonly manages Linux services.

```
systemctl status <service>
journalctl -u <service> -n 100
journalctl -u <service> -f
```

When a service is down, inspect its status and logs before restarting it.

6. Network Troubleshooting

Check interfaces, routes, listening ports, and HTTP connectivity.

```
ip addr
ip route
sudo ss -lntp
curl -I http://localhost:<port>
```

A port being open does not necessarily mean the application is healthy. Check the application response and logs as well.

7. Safe Remediation

SIGTERM is normally preferred for graceful process termination.

```
kill <PID>
kill -9 <PID>
```

Use SIGKILL only when a process does not respond appropriately to graceful termination. Never blindly terminate critical processes such as PID 1.

Recommended workflow: **observe** → **collect evidence** → **identify root cause** → **obtain approval if needed** → **remediate** → **verify**.